

WEEK	DATE	TOPIC	Evaluations
1	Sept 1	Introduction to R and RStudio IDE <i>Students will be introduced to the R programming language and the RStudio integrated development environment (IDE). The session will cover basic navigation, writing and running code, and organizing work within RStudio.</i>	
	Sept 3	R Objects and Data Types <i>Students will learn how R stores information using common object types and data structures. Hands-on activities will introduce variables, vectors, and basic data types used in biological data analysis.</i>	
2	Sept 8	Next Generation Sequencing Lecture & Mini-TBL <i>This session introduces the fundamentals of next-generation sequencing and common genomics data types. Students will also participate in a mini team-based learning based on:</i> <i>“From bench to bytes: a practical guide to RNA sequencing data analysis” by Dawadi et al.</i>	
	Sept 10	No in-person class <i>There will be no in-person class meeting. Students should complete the assigned TBL questions following instructions posted on the course website.</i>	TBL Discussion Questions Due
3	Sept 15	R Data Structures <i>Students will explore common R data structures, including vectors, matrices, lists, and data frames. Activities will focus on accessing, organizing, and manipulating data for downstream analysis.</i>	Quiz #1
	Sept 17	Paper 1: Team-Based Learning (TBL) <i>Students will work collaboratively to critically evaluate an assigned primary research article:</i> <i>“Influenza infection triggers disease in a genetic model of experimental autoimmune encephalomyelitis” by Blackmore et al.</i>	

4	Sept 22	Factors & dplyr (Part I) <i>Students will learn how categorical variables are represented as factors in R and begin using the dplyr package for data manipulation.</i>	TBL Discussion Questions Due
	Sept 24	Manipulating data with dplyr (Part II) <i>Students will continue developing data wrangling skills using dplyr. Topics will include sorting, creating new variables, and summarizing data to prepare datasets for analysis and visualization.</i>	
5	Sept 29	Manipulating data with dplyr (Part III)	Quiz #2
	Oct 1	Paper 2: Team-Based Learning (TBL) <i>Students will collaboratively analyze and answer structured discussion questions on a second primary research article:</i> <i>“RNA-Seq Transcriptome Profiling Identifies CRISPLD2 as a Glucocorticoid Responsive Gene that Modulates Cytokine Function in Airway Smooth Muscle Cells” by Himes et al.</i>	
6	Oct 6	Data Visualization Part I <i>Students will be introduced to principles of effective data visualization using ggplot2. Activities will focus on constructing basic plots and understanding the grammar of graphics.</i>	TBL Discussion Questions Due
	Oct 8	Data Visualization Part II	
7	Oct 13	Data Visualization Part III	Quiz #3
	Oct 15	Paper 3: Team-Based Learning (TBL) <i>Students will collaboratively analyze and answer structured discussion questions on a third primary research article:</i> <i>“Estrogen Receptor Alpha Mutations in Breast Cancer Cells Cause Gene Expression Changes through Constant Activity and Secondary Effects” by Arnesen et al.</i>	
8	Oct 20	R Markdown for Reporting Results <i>Students will learn to create reproducible analysis reports using R Markdown. The session will cover combining code,</i>	TBL Discussion Questions Due

		<i>figures, results, and written interpretation into a single document.</i>	
	Oct 22	RNA-seq Data QC <i>Students will learn the principles of quality control for RNA-seq data and examine common measures used to assess sequencing and read quality.</i>	
9	Oct 27	DE analysis (tumor vs normal samples) <i>Students will apply differential expression analysis to compare gene expression between tumor and normal samples. Activities will focus on interpreting statistical results and identifying biologically meaningful patterns in gene expression data.</i>	
	Oct 29	DE analysis continued (tumor vs normal samples)	Quiz #4
10	Nov 3	Paper 4: Team-Based Learning (TBL) <i>Students will collaboratively analyze and answer structured discussion questions on a fourth primary research article:</i> <i>“Natural genetic variation in wild-derived mice controls host survival and transcriptional responses during endotoxic shock” by Sabikunnahar et al.</i>	
	Nov 5	Project 1 Q&A <i>Students will work on their individual differential expression analysis projects and receive guidance on their analytical approach and interpretation of results. Class time will also provide an opportunity to troubleshoot code and discuss project expectations.</i>	
11	Nov 10	Intro to Quarto <i>Students will be introduced to Quarto as a tool for creating reproducible computational documents and reports. Activities will demonstrate how to integrate code, results, figures, and narrative text into a polished document.</i>	Project 1 Due
	Nov 12	Paper 5: Team-Based Learning (TBL) <i>Students will collaboratively analyze and answer structured discussion questions on the final primary research article.</i>	
12	Nov 17	DNA-Seq Data Basics	TBL Discussion Questions Due

		<i>This exercise introduces the fundamentals of DNA sequencing data and common approaches used to identify genetic variation. Students will explore concepts including sequencing coverage, variant calling, and mutation types.</i>	
	Nov 19	Mutational signatures (SNVS, Indels, base substitutions) <i>Students will explore common types of somatic mutations and approaches used to characterize mutational patterns in cancer genomes. The session will introduce single nucleotide variants, insertions and deletions, base substitutions, and mutational signatures.</i>	
13	Nov 24	Thanksgiving Break	
	Nov 26		
14	Dec 1	Visualizing Microbial Data <i>Students will explore approaches for visualizing microbial and microbiome datasets in R. Activities may include examining community composition, relative abundance, and differences across samples or experimental groups.</i>	
	Dec 3	Visualizing Microbial Data	Quiz #5
15	Dec 8	Project 2 Q&A <i>Students will work on their individual differential expression analysis projects and receive guidance on their analytical approach and interpretation of results. Class time will also provide an opportunity to troubleshoot code and discuss project expectations.</i>	
	Dec 10	Project 2 Final	Project 2 Due